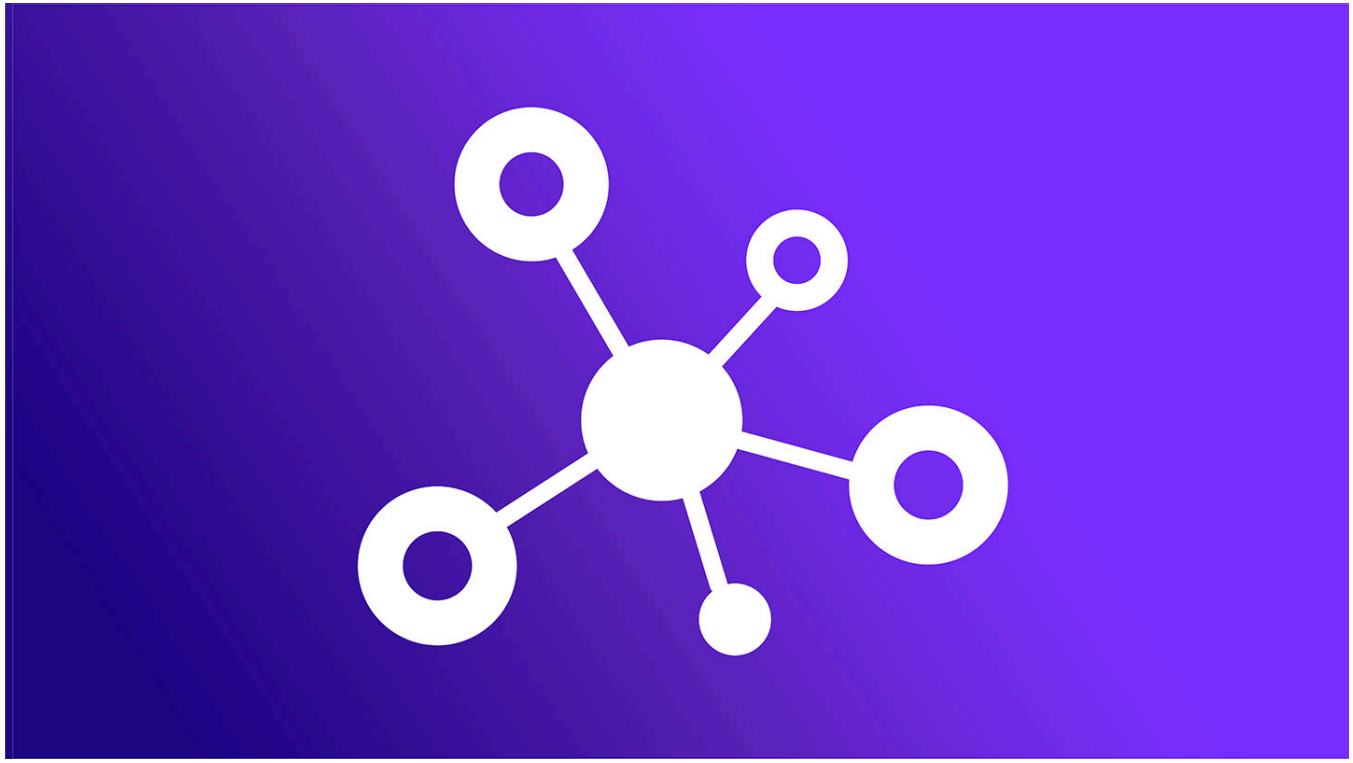
National

Operational Excellence Standard (OES) — Aerial drop installation

HOW22895 | MODIFIED 11/28/2025



Summary

This standard covers the correct installation requirements for all aerial drops. This may refer to either a coaxial or a fiber line from the subscriber premise to the aerial distribution plant.

Quick Links

- What's changed in this article
- At a glance
  - Required tools and hardware
- Aerial drop installation guidelines
- Aerial drop attachment types and descriptions
- Minimum clearance specifications for aerial drops
  - Vertical clearances from other service providers
  - Aerial drop minimum clearances from other structures
- Coaxial attachment at the tap
- Fiber attachments at the optical tap
- Routing of cable/drops
- Splicing of aerial drops
- OES aerial drop installation document owner, contributors, and approvers

What's changed in this article

Date	Summary of change
November 26, 2025	Added two images to **Aerial drop attachment types and descriptions** section.
September 5, 2025	Added to Celestial.
March 21, 2024	Conducted scheduled review and updated.
November 7, 2022	Added guidance for fiber drop installations.
March 8, 2022	Conducted scheduled review and updated.
August 11, 2020	Reviewed for incorporation of Commercial standards.
March 5, 2020	Made minor updates.
February 25, 2019	Released document.

At a glance

This article covers the aerial drop installation OES.

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Required tools and hardware

- Fiber drop hanger — Black CIFA# 96588 — Stainless steel CIFA# 96590

Aerial drop installation guidelines

This OES will cover all requirements for the proper installation of an aerial drop. Within this section the terms “cable, cables or aerial drop” may refer to either a coaxial or a fiber drop. When required, the use of each term will be defined based on the specification.

This document does not substitute for any local, state, or federal code or requirements, but should be used as a supplement to these.

- **Deviations**
  - Practices and configurations outlined in this document are the Comcast standard and may not be modified in their implementation. Any deviations from the

standards prescribed in this document shall be approved by Comcast Corporate Technical Operations.

- **Safety**
  - All personnel are required to always comply with Comcast's security and safety regulations.

- **Workmanship**
  - All work performed will be of the highest possible workmanship. All work will adhere to Comcast's Standards, Local, State, and Federal codes/requirements and be fully compliant to any stipulations made by the Landlord/Agent in control of the property.

- All aerial drops must meet the requirements of Article 820 of the National Electrical Code (NEC) at a minimum.
- Cables used for coaxial aerial installations must have an integral self-supporting strength member ("messengered" cable).
- Cables used for coaxial aerial installations must have a non-flowing flooding compound on the braid wires. Use of alternate dry corrosion resistant treatments must receive prior approval by Comcast Corporate Engineering.
- Series-6 drop cable must be used for all coaxial aerial drops shorter than 150 feet long.
- The use of series-11 or QR-320 cable is permissible only for aerial coaxial drops longer than 150 feet or in situations where its use will improve RF forward signal levels to within acceptable ranges and will therefore eliminate the need for RF signal amplification.
- Drops using series-11 cable or QR-320 cable shall not exceed 300 feet in length.
- Any situation that will require coaxial cable lengths in excess of what is defined must be approved by National Engineering.
- Messenger must be removed from all aerial COAX drop cable from the tap to the first attachment point (span clamp) and from the last attachment point (P-hook) to the Bond block.
- Aerial fiber drop shall be a cable with one fiber.

- Aerial fiber drop will be available in specified lengths preconnectorized; it may also be purchased in a hardened optical connector to stub version.
- When utilizing preconnectorized drops, select a length that will not require excessive amount of fiber to be stored.
- Optical underground cable must include a toneable or traceable component for locating purposes.
- Optical underground cable contains one fiber and can be a pushable round fiber or a flat style fiber.

Aerial drop attachment types and descriptions

New aerial coaxial drop installations:

- Require the use of approved drop clamps designed for dead-ending messengered drop cable for attaching drops.

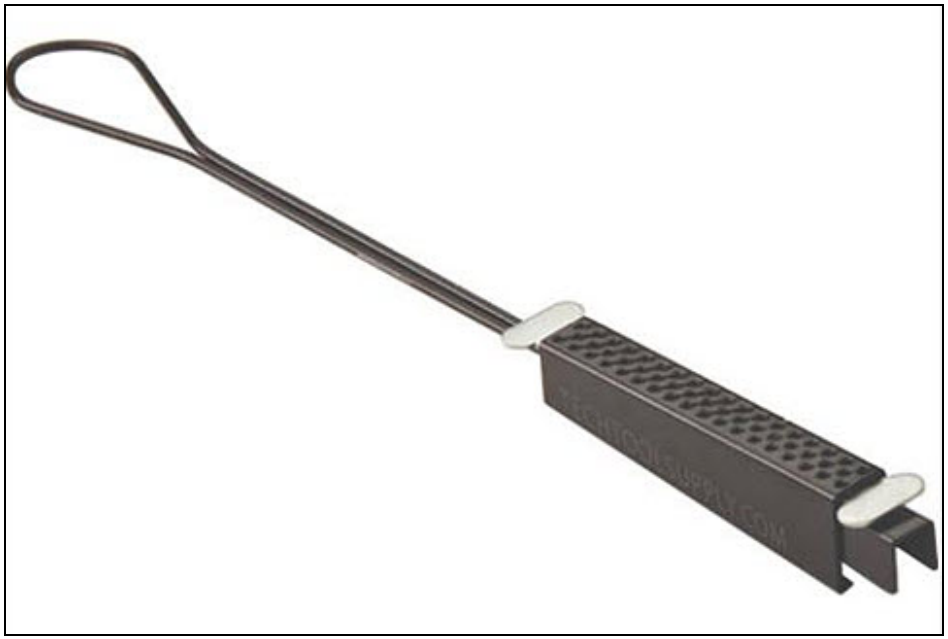
Existing coaxial drops:

- Are acceptable if attached with a properly applied 2-4-4 messenger configuration.
- If the 2-4-4 messenger configuration is not properly applied, it must be replaced with an approved drop clamp.

For all fiber drop installations:

- Approved drop clamps designed to support the fiber drop configuration must be used.

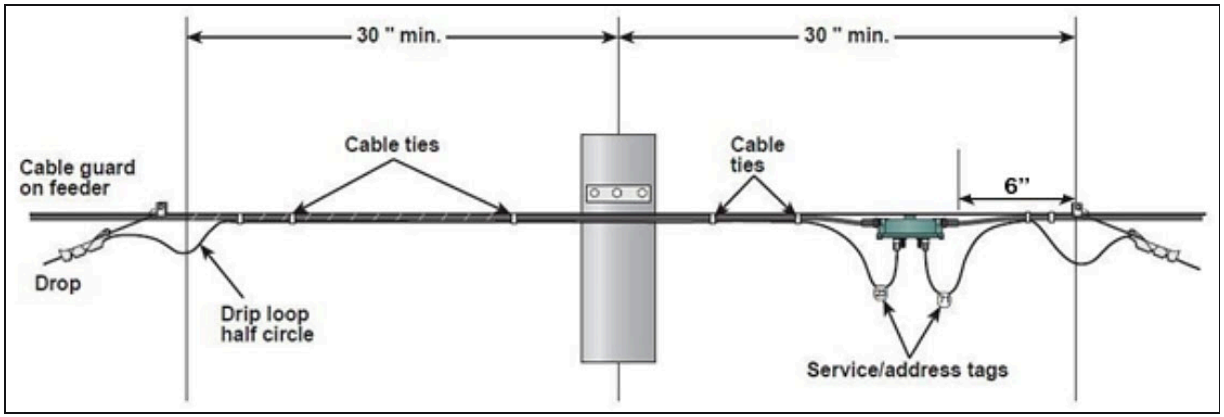




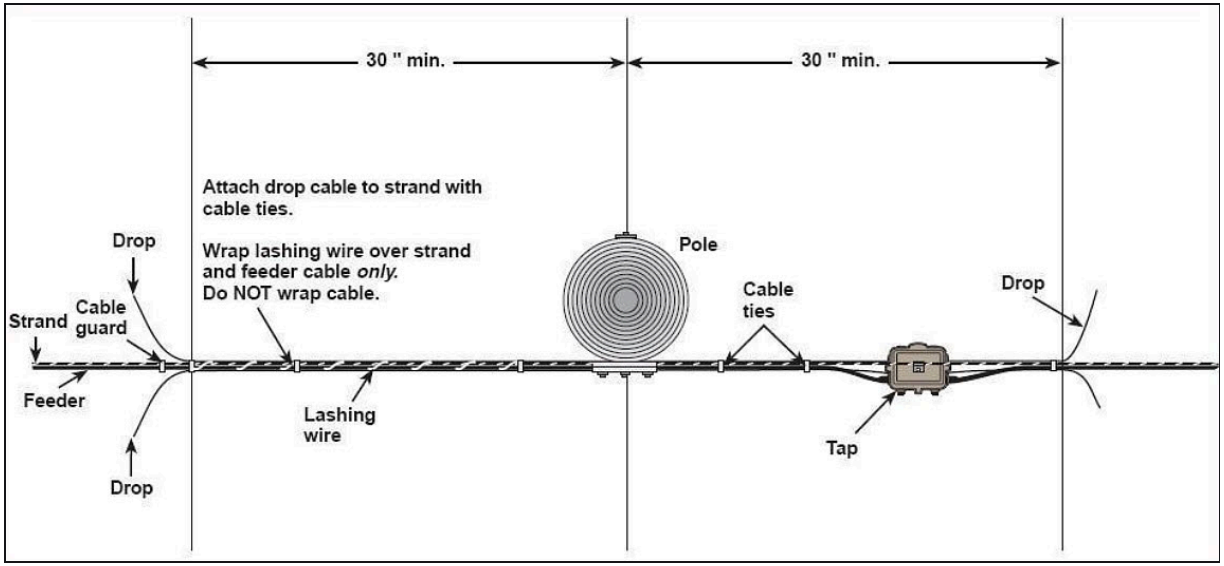
- Span clamps designed for use with a drop clamp must be used and only be attached to the strand.
  - Span clamps must **never** be placed over the lashing wire.
  - Up to four drop clamps may be attached to one span clamp.
- To prevent cable damage, cable guards must be used over hardline cable wherever the hardline cable is lashed up to the strand close to the span clamp location.
- Use these guidelines for using a J hook or a P hook as an attachment on a pole:
  - Use **only** when clearance requirements cannot be met by attaching to the strand. See the Aerial drop minimum clearances from other structures section/table.
  - The J or P hook placement must follow drop clearances from other providers as defined in the Vertical clearances from other service providers table.
  - Supervisor approval is needed before installing a J or P hook to any utility pole unless it is a sub or spot pole.
- Drop cables must **never** be attached to any hardware not issued or installed by Comcast.

Minimum clearance specifications for aerial drops

- The drop cable must be tensioned to match the sag of the power drop.
  - If there is no power drop reference, the drop cable must be tensioned for a minimum of six inches of sag for every 50 feet in length.



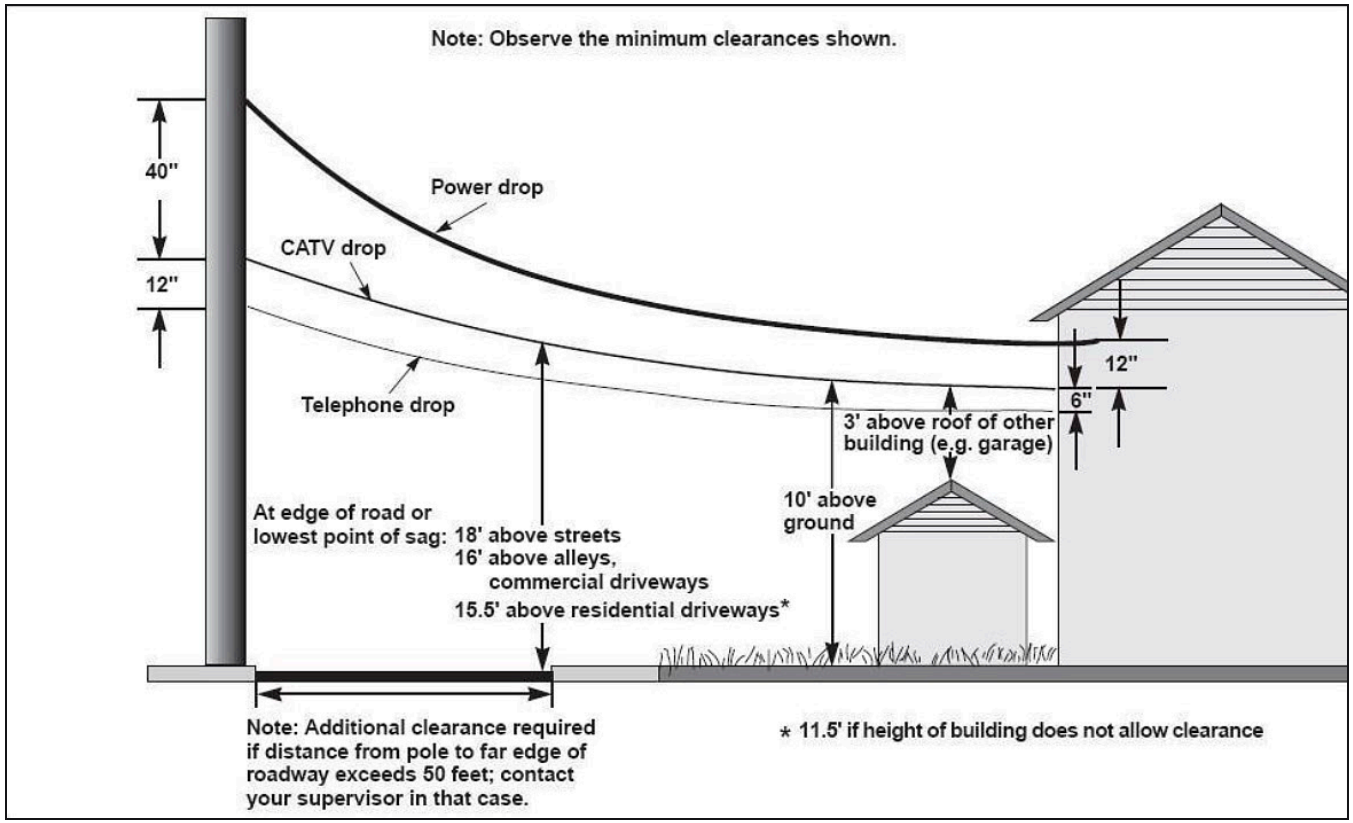
  - A minimum 30-inch square climbing space from the center of the pole must always be maintained.



- Vertical clearances from other services must be maintained as indicated in the **Vertical clearances from other service providers** table below.
- Minimum clearances above roads, driveways, walkways, and other structures must be maintained as indicated in the **Aerial drop minimum clearances from other structures** table below and illustrated diagram above.

Vertical clearances from other service providers

Service	At the pole	Mid-span	At the premise
Clearance from power	40 inches	24 inches	12 inches
Clearance from phone or other telecommunications providers	12 inches	6 inches	6 inches



Aerial drop minimum clearances from other structures

Location	Clearance
Over roads and streets	18 feet
Commercial driveways, parking lots, alleys, and other areas subject to truck traffic	16 feet

Location	Clearance
Over residential driveways that are subject to truck traffic	15.5 feet
Over residential driveways that are not subject to truck traffic or where attachment to the structure does not permit service drops	11.5 feet
Above sidewalks or other pedestrian traffic areas	10 feet
Above roofs that are not accessible to pedestrians	3 feet
All points above a flat roof	10.5 feet

Coaxial attachment at the tap

- When connecting drops to a tap; F-connector, trap, weatherproofing, termination, and tightening guidelines must be followed.
- A drop address tag(s) must be placed on every aerial drop. Each customer's drop requires a tag(s) which indicates their address.
- To accommodate future attachment of the aerial drop to the tap port, or the replacement of the drop fitting, the initial drip loop at the tap should be between 6-7 inches.
    - Under no circumstances should the coax cable be looped 360 degrees.
        - Any 360-degree loops must be removed from any existing coax drops.

    - A minimum diameter drip loop must be present at both messenger attachment points and tap attachment point.
- When routing drop cable from tap to drop clamp attachment point, the cable must follow the path of the strand. See the proper climbing space clearance and with drop routing and attachment to strand diagram

  .
    - Tie-wraps or other cable securing devices shall be used to secure the drop cable to the strand every 18-24 inches as needed.
    - Tie-wraps must be installed in a manner that does not negatively impact the cable's characteristic impedance.
    - The drop cable must be secured underneath the strand to deter rodent damage.

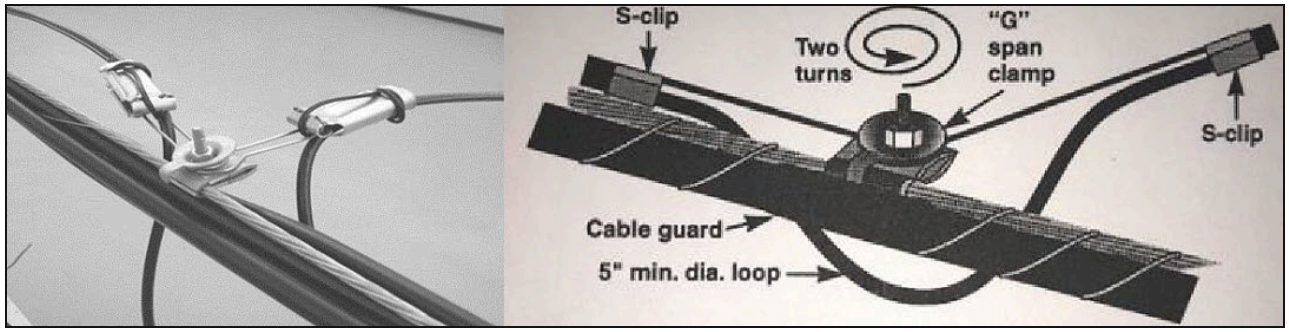
Fiber attachments at the optical tap

- When connecting fiber drops to an optical tap; hardened SC/APC optical connections, proper weatherproofing, termination, and tightening guidelines must be followed.
- One 12-inch diameter slack loop must be left from the drop cable strand grip to the aerial optical tap port. Secure the slack loop with cable ties.
    - **Caution:** Do not place drip loops in the fiber optic cable during installation. Forming a drip loop in the cable will most likely exceed the bend radius or break the cable's strength members, causing fiber damage and attenuation.
- A drop address tag(s) must be placed on every aerial drop. Each customer's drop requires a tag(s) which indicates their address.
- Excess drop storage shall be maintained at the subscribers' premise enclosure.
- A minimum of 20 feet is recommended, not to exceed the manufacturer's recommendations for the enclosure.
- For flat drop, the associated jacket and strength members must be removed for proper storage in the premise enclosure.

- Only 2–4 inches of the strength members shall be present in the premise enclosure.
- Follow the manufacturers' recommendations and/ or tool(s) for the removal of these materials.

Mid-span attachment specifications and description

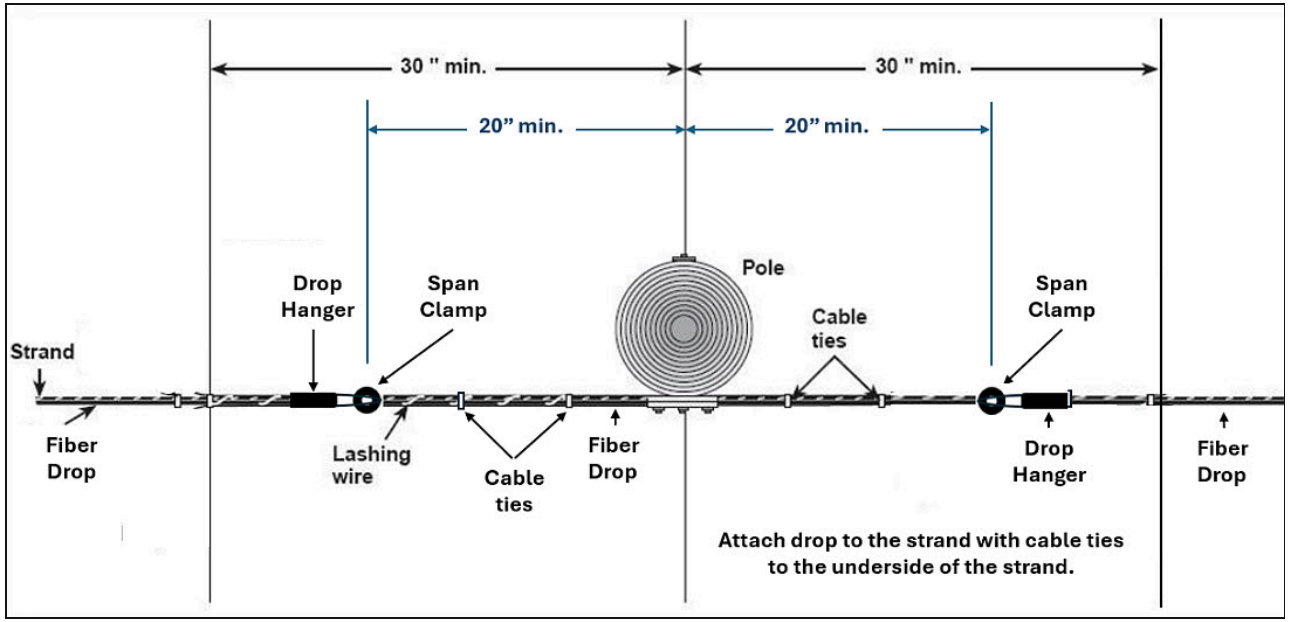
- Mid-span attachments must **always** be installed from the pole to the mid-span point, and then to the premise and removed in the reverse.
- Coaxial messengered drop cable must be attached to the span clamp using drop hangers or the messenger wrap method with S clips. S clips must be installed by hand only, do not use a tool as this could damage the cable.
  - S clips shall be attached no greater than six inches from the stand attachment point.
  - Existing mid-span drops attached with a properly applied 2-4-4 messenger configuration is acceptable.
    - If the 2-4-4 messenger configuration is not properly applied, it must be replaced with an approved drop clamp.
- Fiber mid-span attachment must be made utilizing two drop hangers with 12 inches of excess provided at the attachment.



- The drop sag between mid-span attachment points shall not exceed six inches per 50 feet.

Jump pole — Pole-to-pole attachment utility right of way and fiber

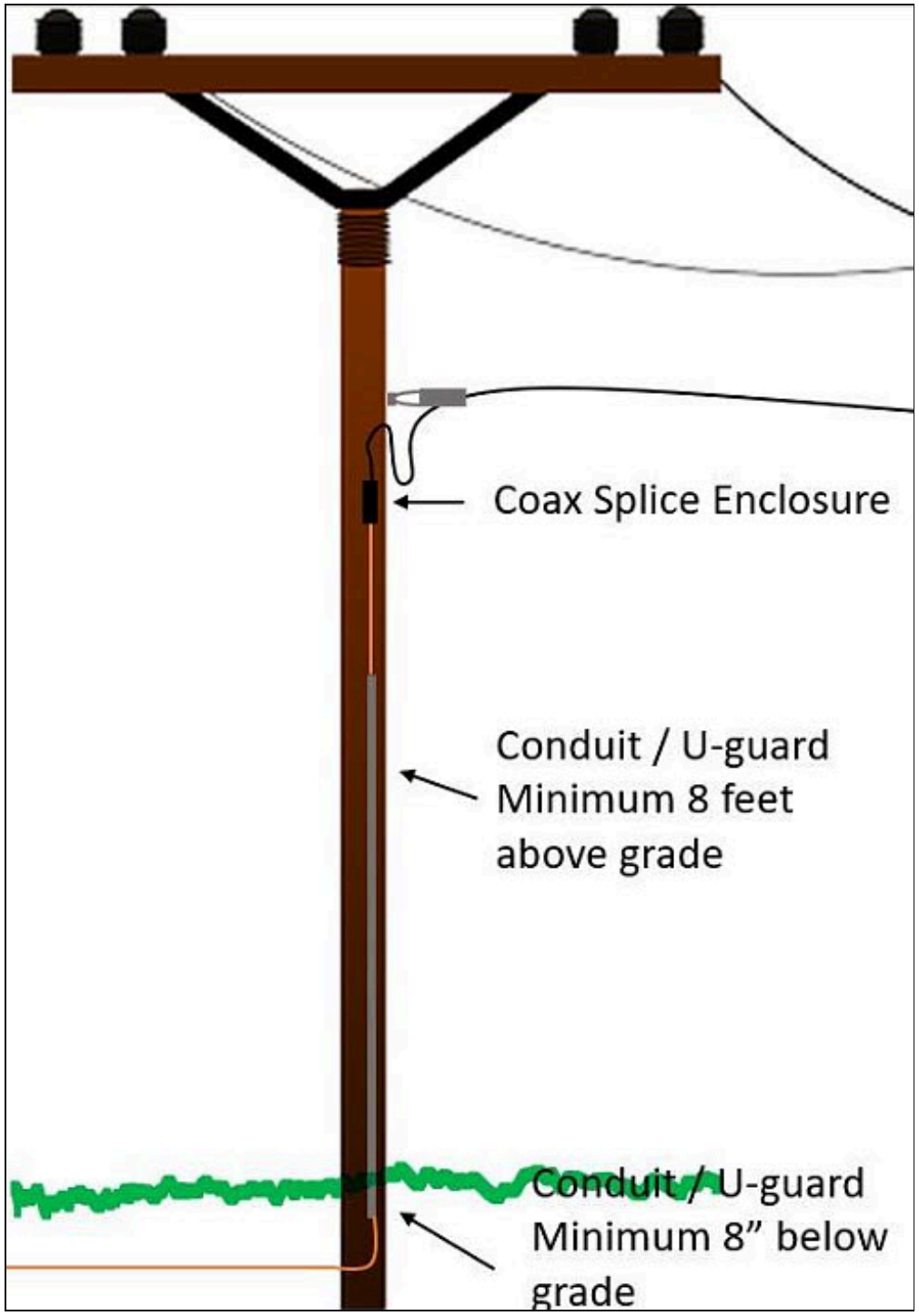
- All fiber drop attachments from pole-to-pole require the drop connection to be made to the strand with a span clamp.
  - The pole-to-pole connection shall not be attached directly to the pole.
- Two approved drop clamps must be used to attach the fiber drop from span clamp to span clamp.
- The drop sag between pole attachment points shall not exceed six inches per 50 feet.
- See figure below for approved clearances and attachment locations.



Aerial to underground drop specifications

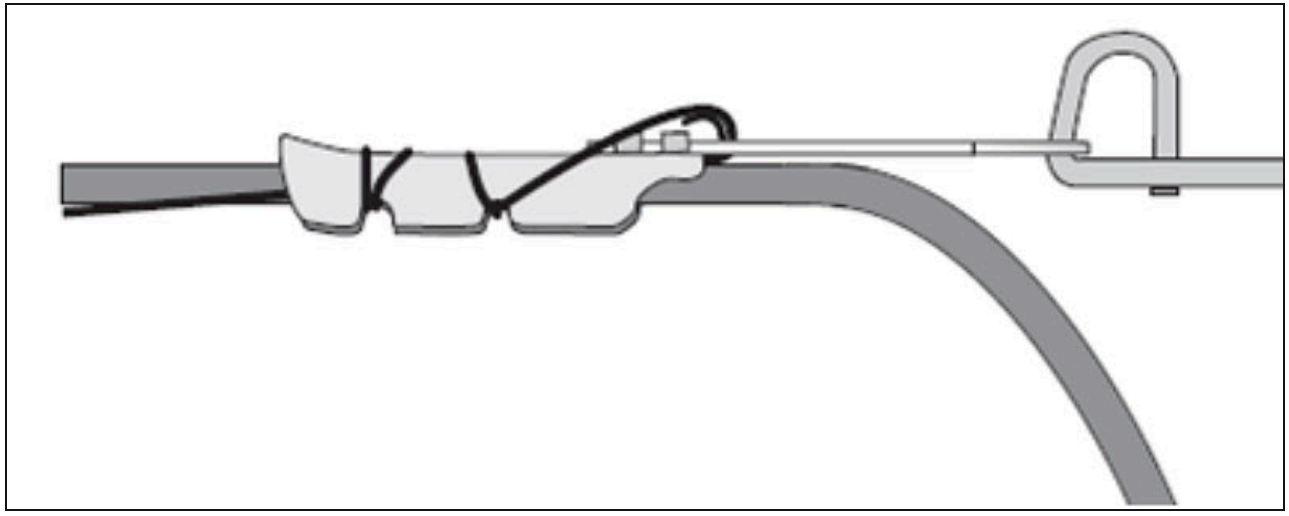
- **Coax:** When the tap is at the pole where you are going from aerial to underground:
  - The drop must be an underground cable, and it must be one continuous piece from the tap to the bond block at the premise service distribution point.
- When the drop runs from an aerial tap to a sub pole and then underground to the subscriber's premise, the drop from the tap to the sub pole shall be an aerial cable with a splice transitioning the cable from Aerial to underground at the sub pole.

    - The splice should be positioned, so it’s not vertical or at the bottom of the drip loop. If the splice is vertical, it shall be in an approved encapsulating device.
- **Fiber:** When the fiber drop runs from aerial to underground or underground to aerial the drop should be one continuous fiber drop with the underground portion being in conduit or micro duct.
- One transition splice from aerial to underground fiber at the pole is acceptable. The transition splice shall be located above the eight feet of conduit or U guard at the pole.
- In all cases, the drop must be protected with conduit or U guard. This must be from eight inches below ground to eight feet above the ground at the pole and from eight inches below ground to the bottom of the service distribution enclosure at the premise.


Coax Splice Enclosure
Conduit / U-guard
Minimum 8 feet
above grade
Conduit / U-guard
Minimum 8” below
grade

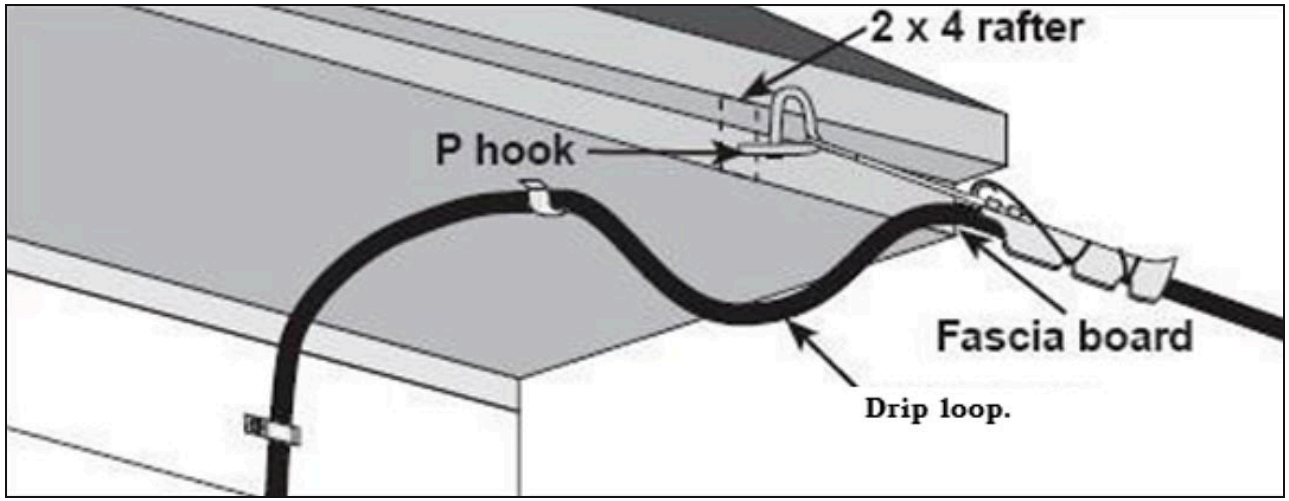
Attachment at the premise for aerial drops

- The messenger drop cable must be attached to the drop clamp and P hook per manufacturer's installation instructions.
- Existing coaxial drops attached with a properly applied 2-4-4 messenger configuration is acceptable.
  - If the 2-4-4 messenger configuration is not properly applied, it must be replaced with an approved drop clamp.
- Aerial service drops must **never** be attached to the power service mast.
  - A separate Telecom mast may be used if provided.
- Aerial service drops must **never** be attached to TV antennas, chimneys, lightning rods, or gutters.



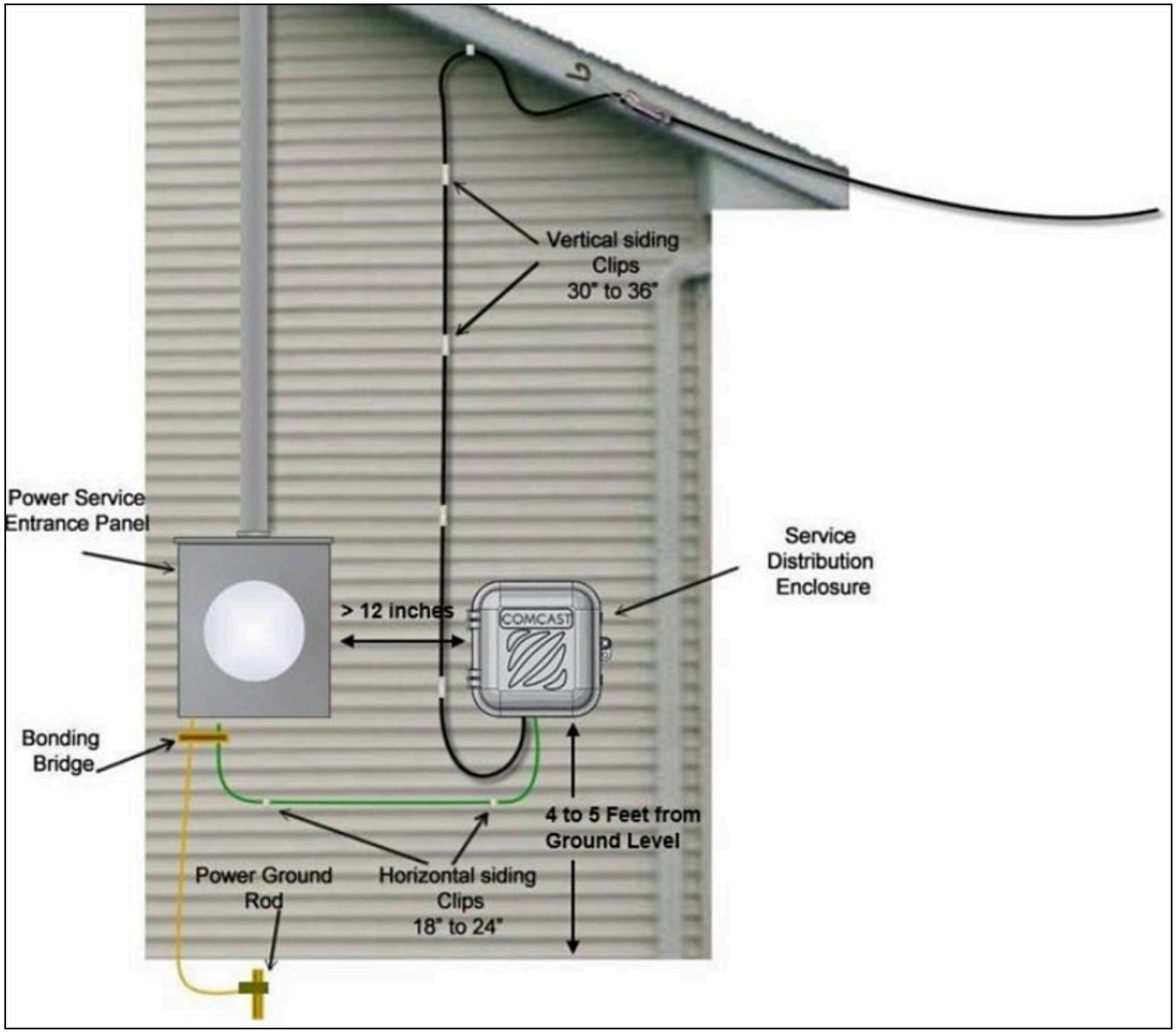
- P hooks must **always** be installed into a stud/rafter or other suitable support material, with the drop's weight pulling perpendicular to the hook's direction of insertion, or as

obliquely from it as is possible, to prevent the hook from being pulled out.



- When attaching to brick or masonry, the P hook must be used with an approved anchor.
- **For coaxial cable:** A minimum 5-inch-diameter less than 270-degree drip loop must be made before fastening or clipping the cable to the premise exterior.
- **For fiber cable:** One 12-inch storage loop shall be made at the P hook attachment.
- The cable must **never** be attached to water pipes, gas lines, electrical power lines, electrical power masts, electrical power conduit, or any other object to which its attachment might interfere with other services or utilities.
- The cable must **never** be attached to rain gutters or downspouts.
- The cable must be routed and fastened to always maintain the minimum bend radius of the cable.
- The cable must be fastened or clipped every 18 to 24 inches if routed horizontally and fastened or clipped every 30 to 36 inches if routed vertically.
- When routing cables on aluminum or vinyl siding, the appropriate vertical or horizontal siding clips must always be used as to not make holes in the siding.
- Aerial flat fiber drops with strength members must have them removed and the associated jacket for proper storage in the premise enclosure.
    - Only 2–4 inches of the strength members shall be present in the premise enclosure.

- Follow the manufacturer's recommendations and/ or tool(s) for the removal of these materials.



Routing of cable/drops

- Aerial drops shall not be in contact with any tree branches or other obstructions and shall maintain safe vertical clearances.
- Aerial drops shall not interfere with climbing space at the pole.
- Aerial drop routing shall follow the power service drop wherever possible.
- The aerial drop cable shall **never** cross property lines outside of the municipal right-of- way and should always avoid aerial cable trespass.
    - A mid-span drop attachment should be used to avoid aerial cable trespass.
- The aerial drop cable shall not cross over any swimming pool.
- The aerial drop cable shall not cross over a utility line.
- The aerial drop cable must **never** be tied to the top side of the strand or pole hardware to prevent damage by animals or falling debris.
- The drop cable must **never** be run around the back side of the pole or left hanging loose.

Splicing of aerial drops

- All coaxial aerial drops shall not have any splices between the tap and the grounding point at the premise.
    - An aerial to underground drop with a sub / spot pole would be the only exception. See the aerial to underground drop illustrated diagram.
- Fiber aerial drops shall not have any splices between the optical tap and the premise enclosure.
    - A damaged fiber aerial drop must be replaced.
    - An exception may be made during emergency restoration; aerial fiber drops may be spliced, and a new drop replacement must be scheduled within 14 days.

OES aerial drop installation document owner, contributors, and approvers

Document owner:

- James Kolcun, Technical Operations, HQ

Contributors:

- Scott Melanson, Technical Operations, Northeast Division
- Mike Darnell, Technical Operations, Central Division
- James Mascarenas, Technical Operations, West Division

Approvals:

- Meena Soleiman, SVP Tech Ops & Eng: Northeastern Division
- Rachel Solomon, SVP Tech Ops & Eng: Western Division
- Tony Speller, SVP Tech Ops & Eng: Central Division
- Stephen Zanetich, VP Technical Operations: HQ
- Patrick McDonald, VP2 Technical Operations: HQ
- Michael Hewitt, SVP Technical Operations: HQ

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